

Introduction to

Machine Learning

“Black Box”

Features, Labels

Examples

Model

Training

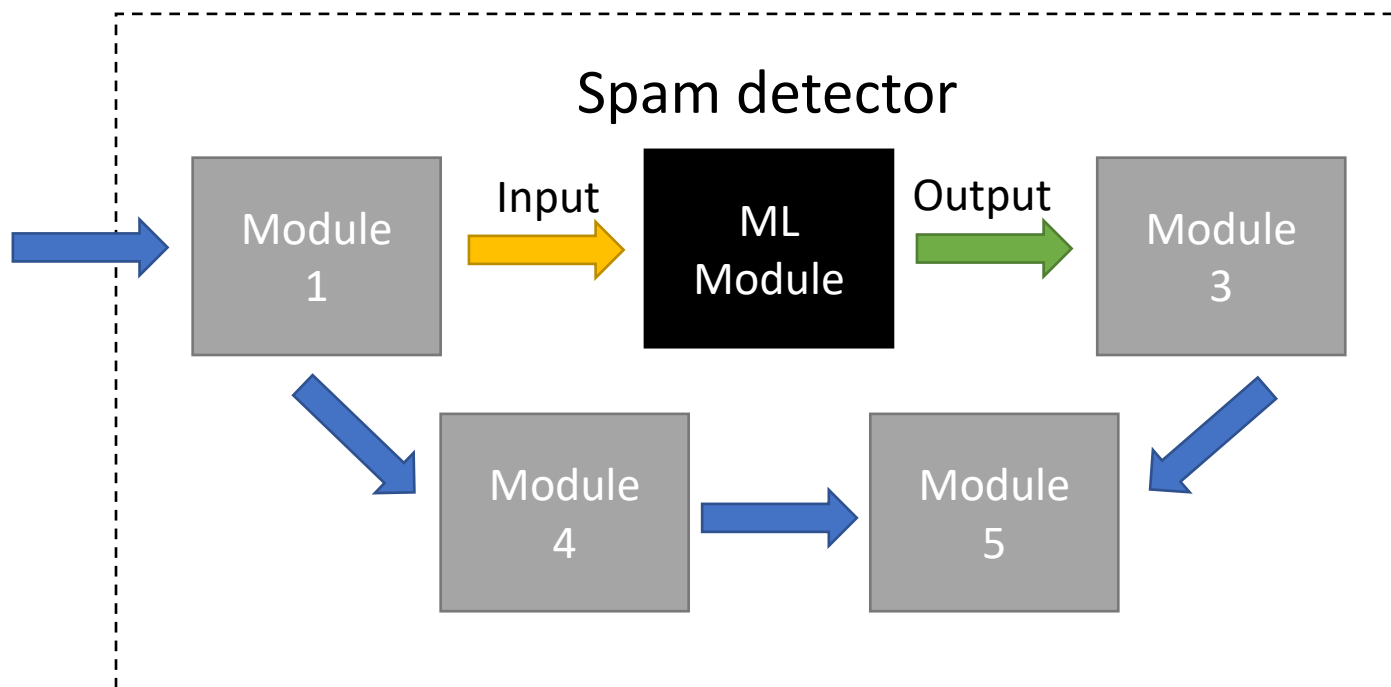
Under fitting

Over fitting

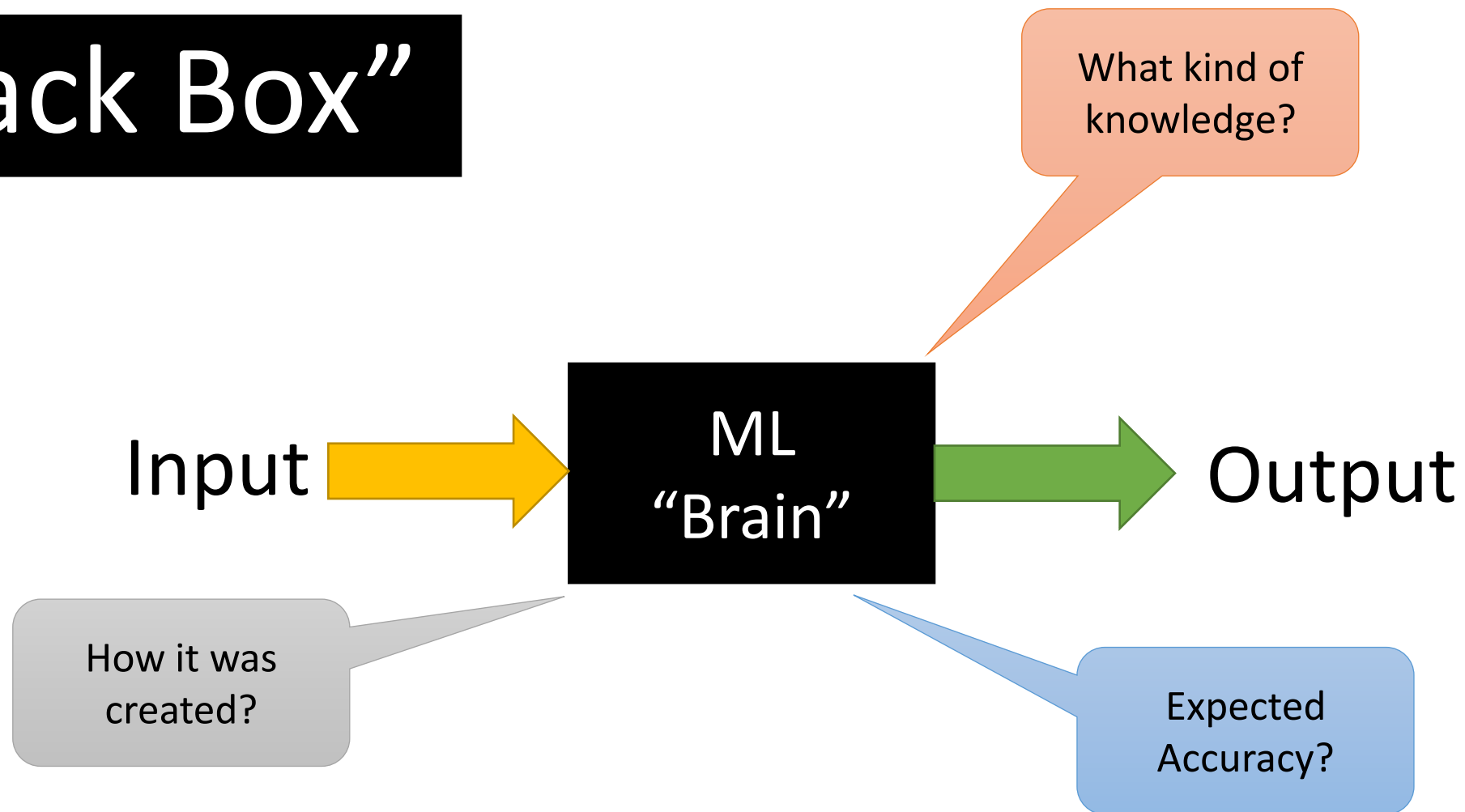
“Black Box”

Email System

Incoming Emails



“Black Box”



Features and Labels

(feature 1, feature 2,...,feature n)

Input



ML
"Brain"



Output

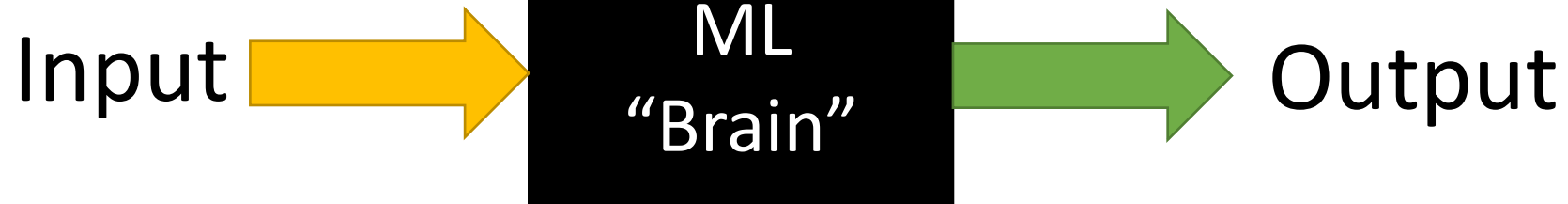
Size
Number of bedrooms
Street
Area
Overall Condition
Year Built
....

Market house price



Features and Labels

(feature 1, feature 2,...,feature n)



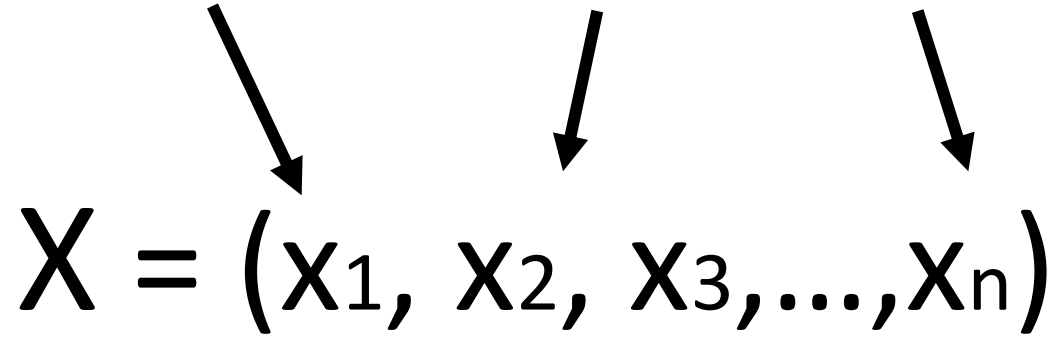
Sender's address
Hour of the day the email was sent
Number of "spam" words - "free", "money"
Number of capitalized words
....

Spam Detector



Features and Labels

(feature 1, feature 2,...,feature n)



The diagram illustrates the mapping from a list of feature names to a feature vector. Three arrows point from the words 'feature 1', 'feature 2', and 'feature n' in the text above to the corresponding elements x_1 , x_2 , and x_n in the vector X below.

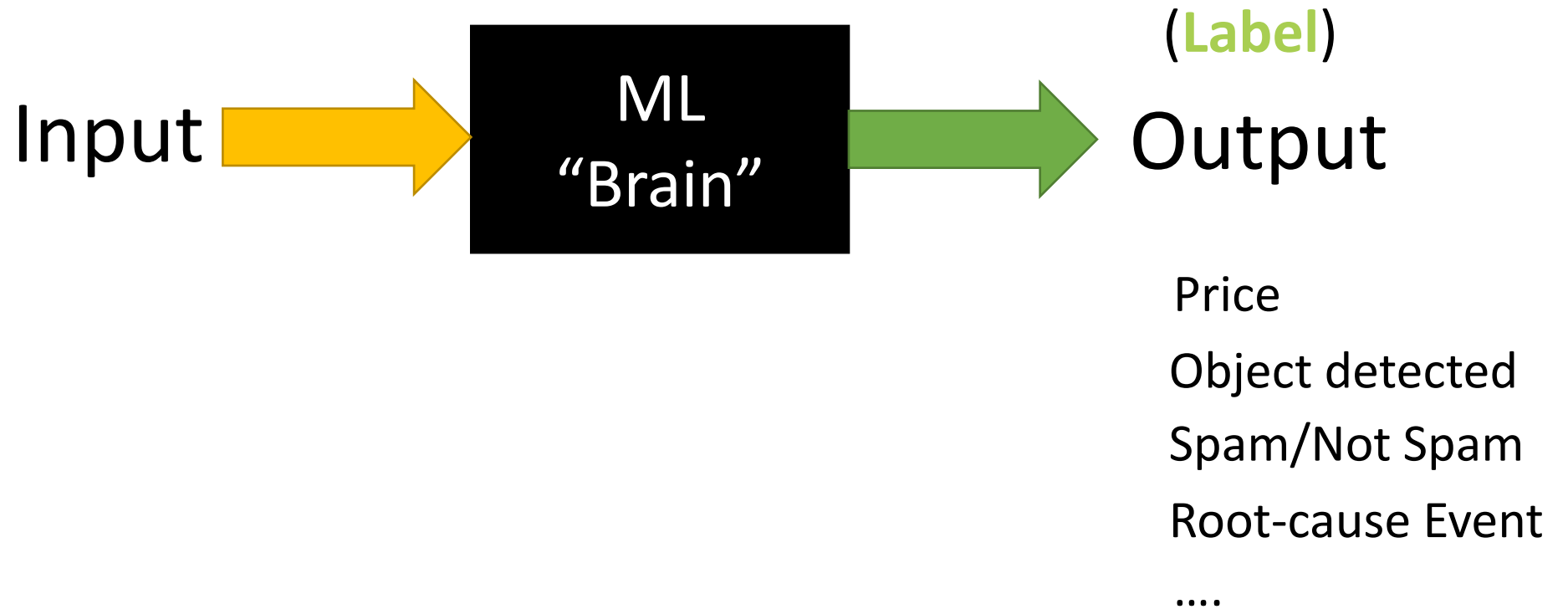
$$X = (x_1, x_2, x_3, \dots, x_n)$$

Features and Labels

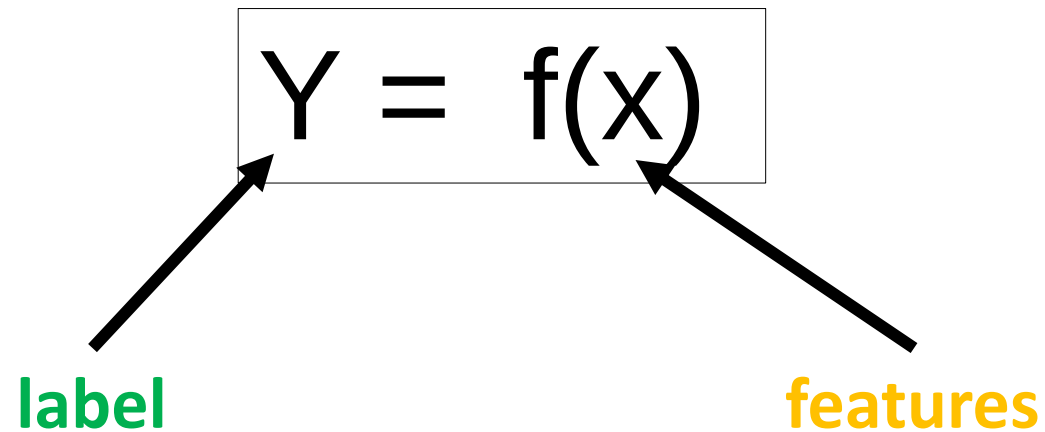
“Garbage in, garbage out”

Features selection

Features and Labels



Features and Labels



Examples

Labeled example

(x, y)

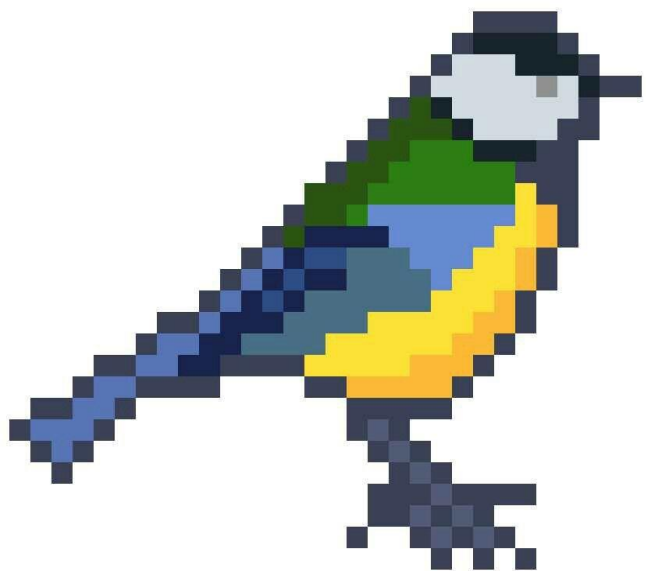
(features, label)

Unlabeled example

$(x, ?)$

(features, ?)

Examples



Features →

Label →

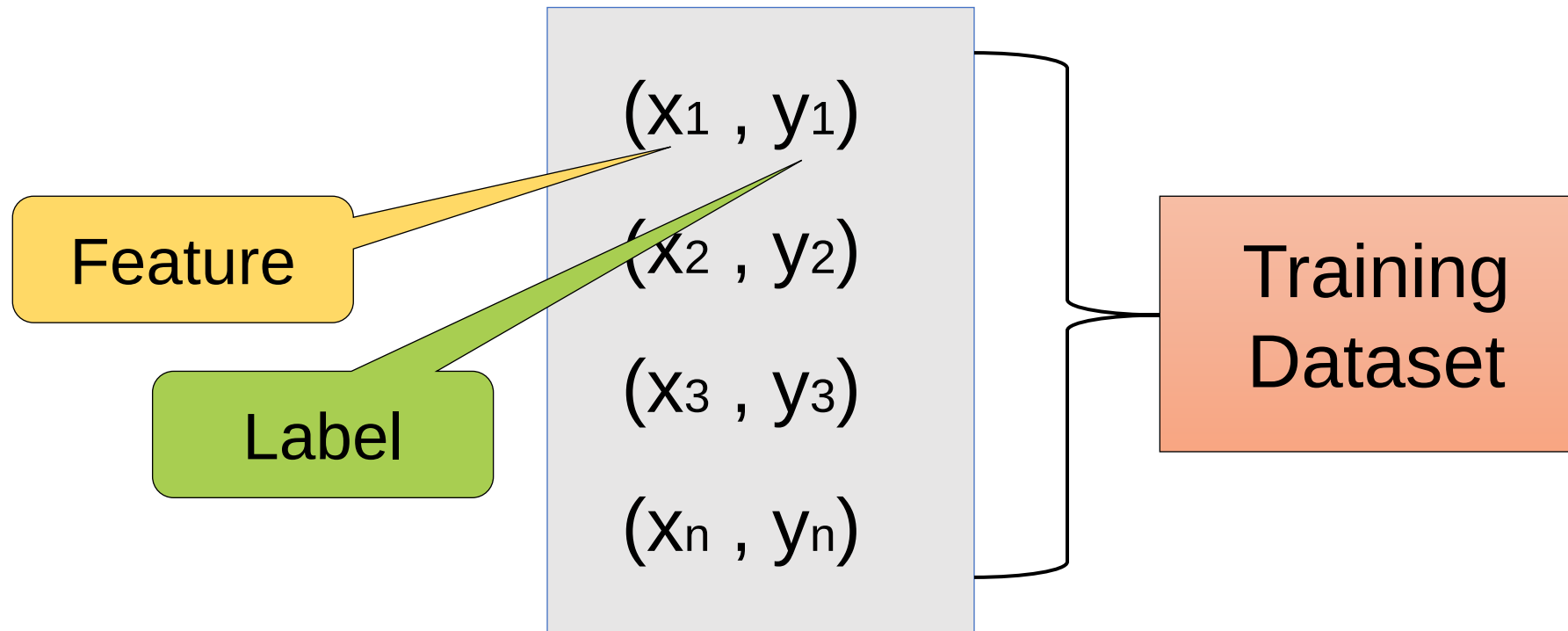
Pixels....



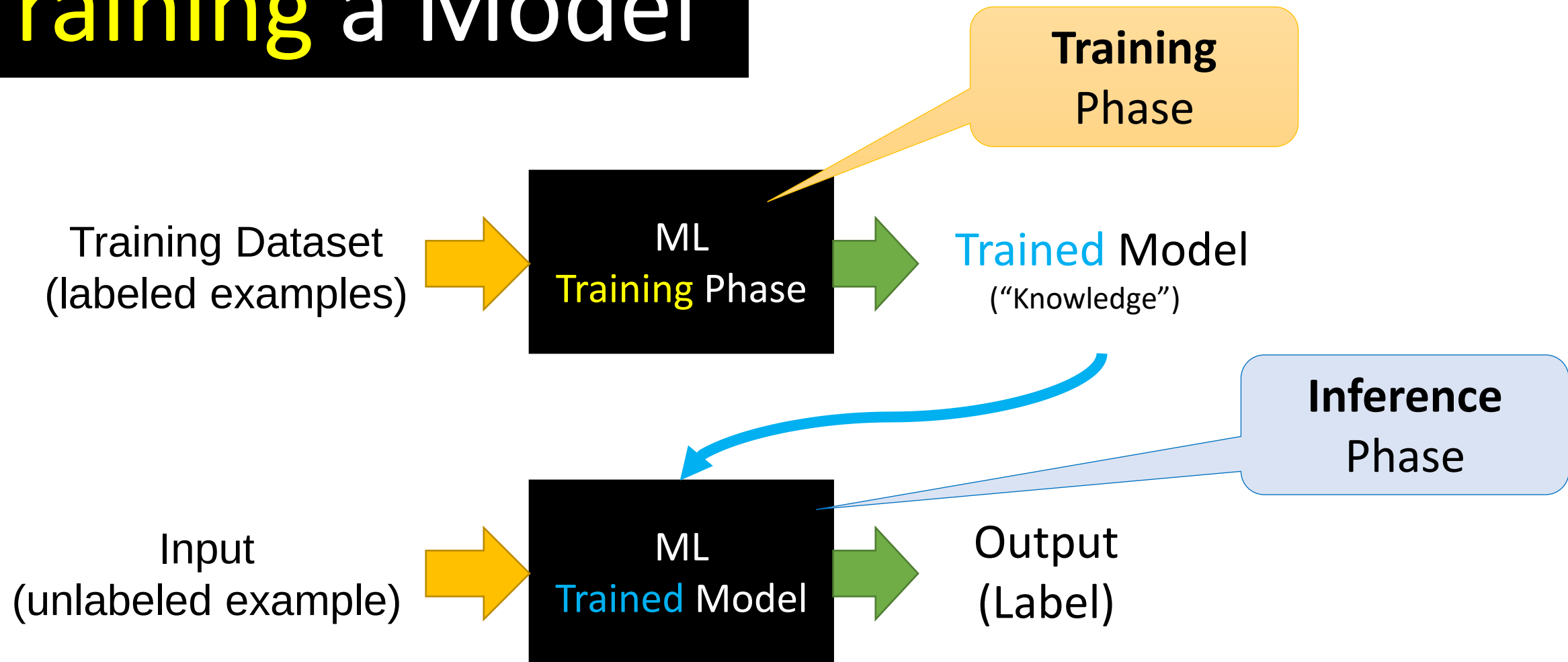
Object = "Bird"

Examples

Labeled Examples



Training a Model



Training a Model



**“This is a
toucan!”**

Training a Model



Dog



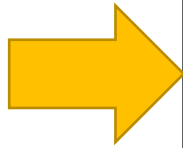
Lion



Zebra

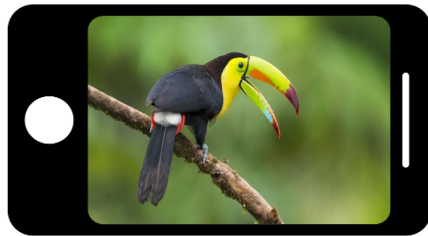


Koala



**Trained
Model**

Model Snapshot



Toucan!

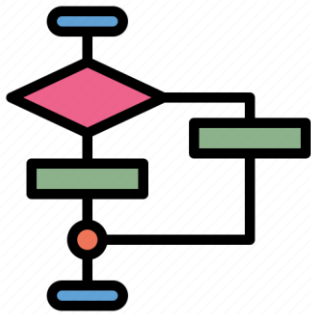
Aiming for Generalization

Overview

Training a model is
NOT an easy task...

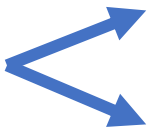


Optimization algorithm
using the **TRAINING** dataset



Objective: making a good prediction on **NEW** data

Generalization

Challenges  Under-fitting
Over-fitting

Aiming for Generalization

Under-fitting

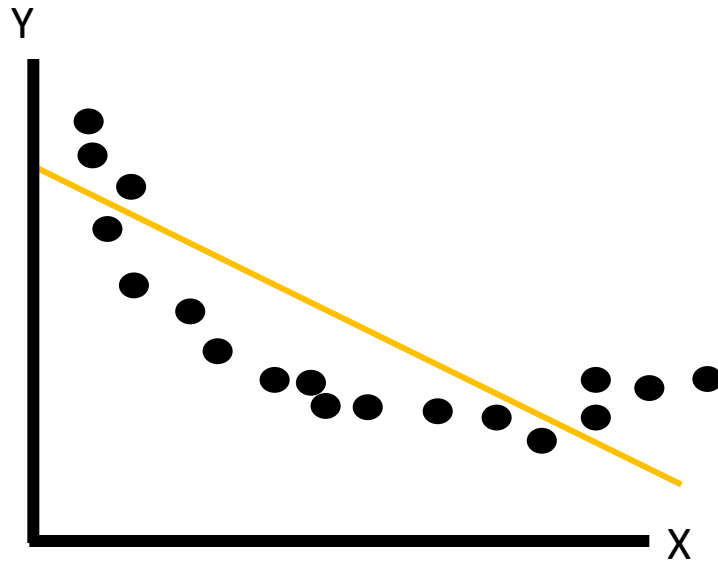
The trained model is **not** working well on the **training** data and can't generalize to **new** data



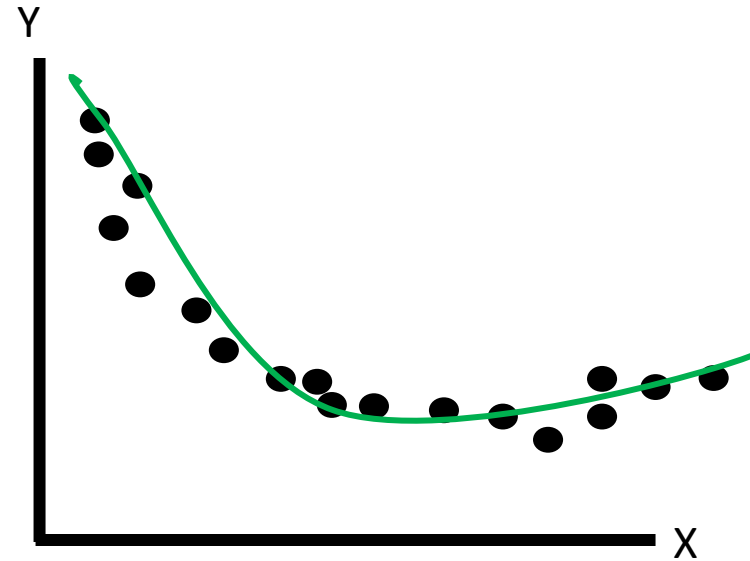
Aiming for Generalization

Under-fitting

Under-fitting

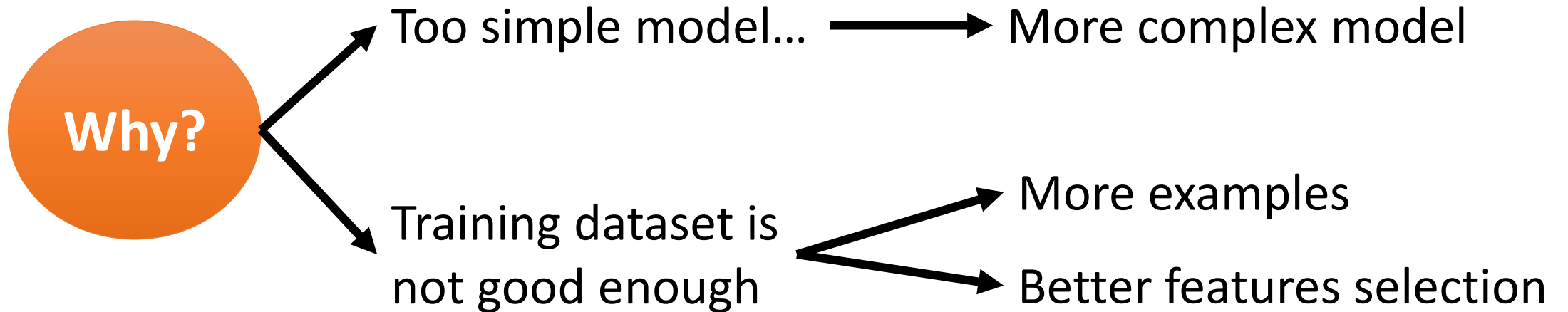


Good fit



Aiming for Generalization

Under-fitting



Aiming for Generalization

Under-fitting

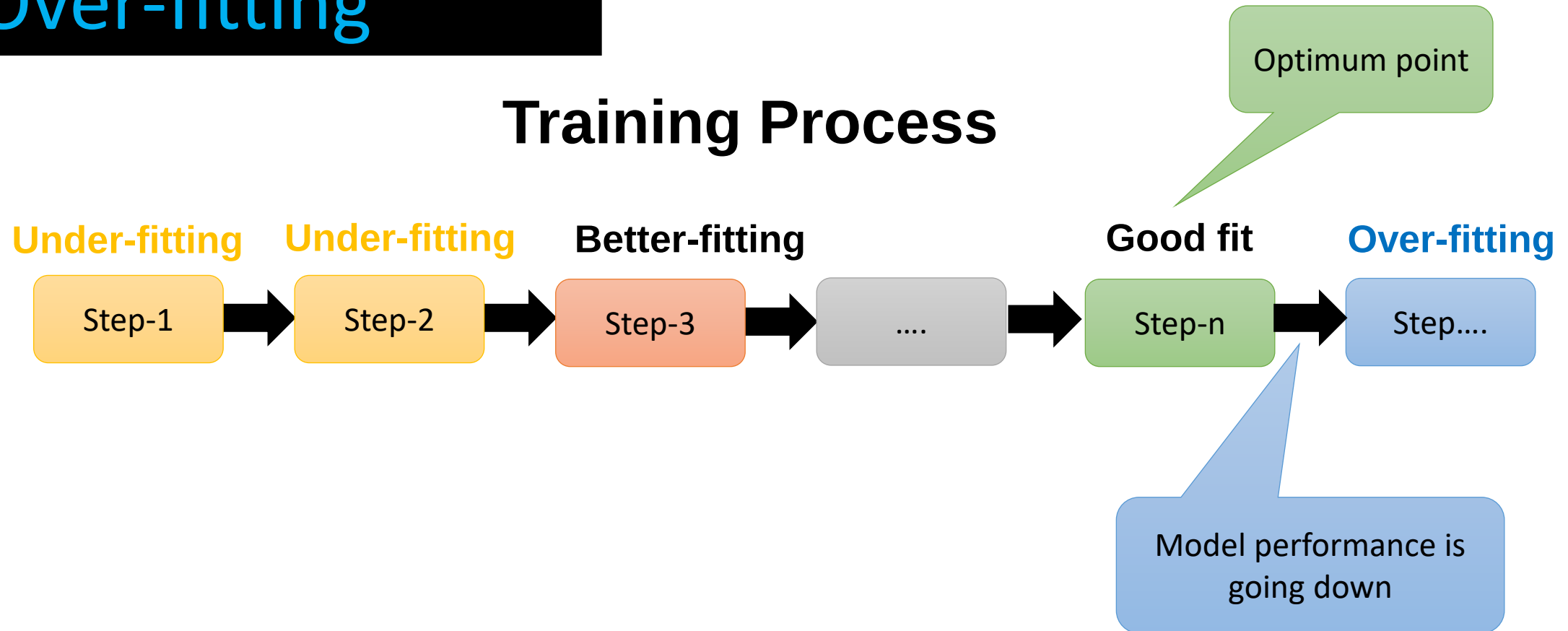
Training Process



Aiming for Generalization

Over-fitting

Training Process



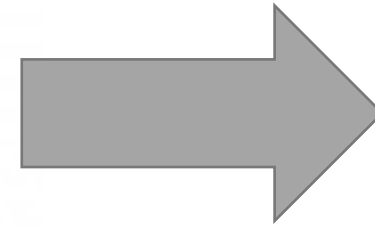
Aiming for Generalization

Over-fitting



Over-Generalized

~20 items



Supermarkets here
are **VERY** cheap!



Aiming for Generalization

Over-fitting

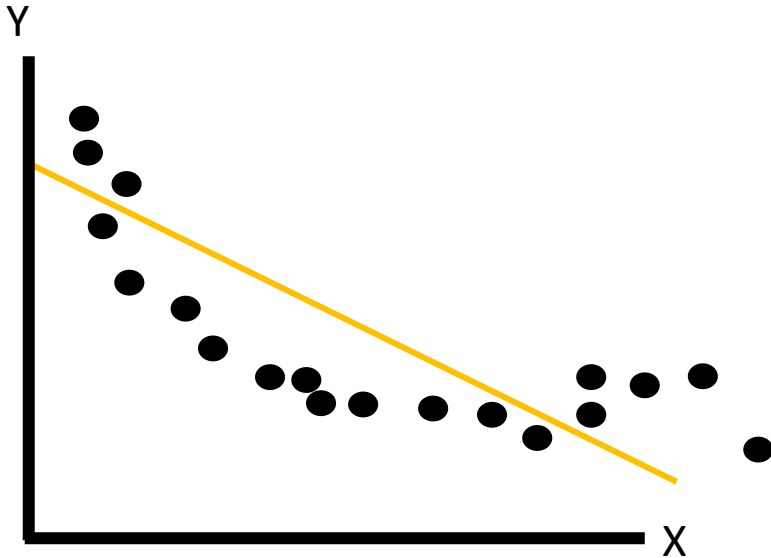
The trained model is working very well on the **training** data and can't generalize well to **new** data



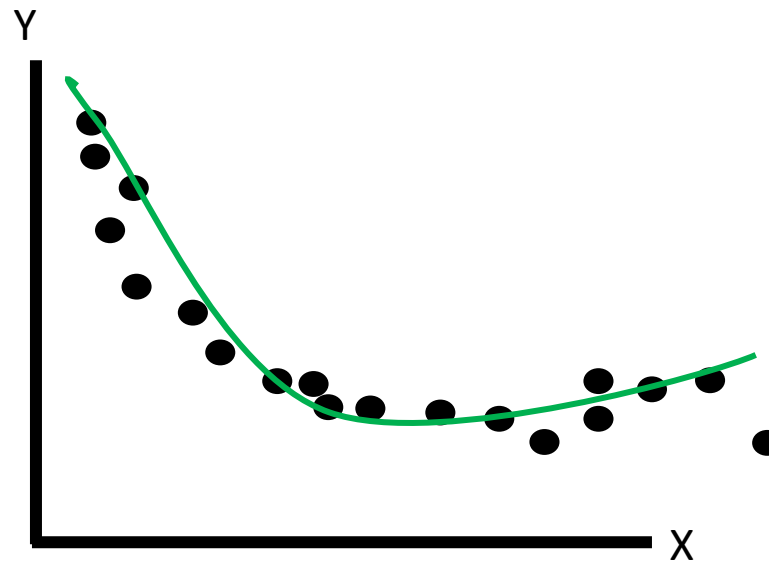
Aiming for Generalization

Over-fitting

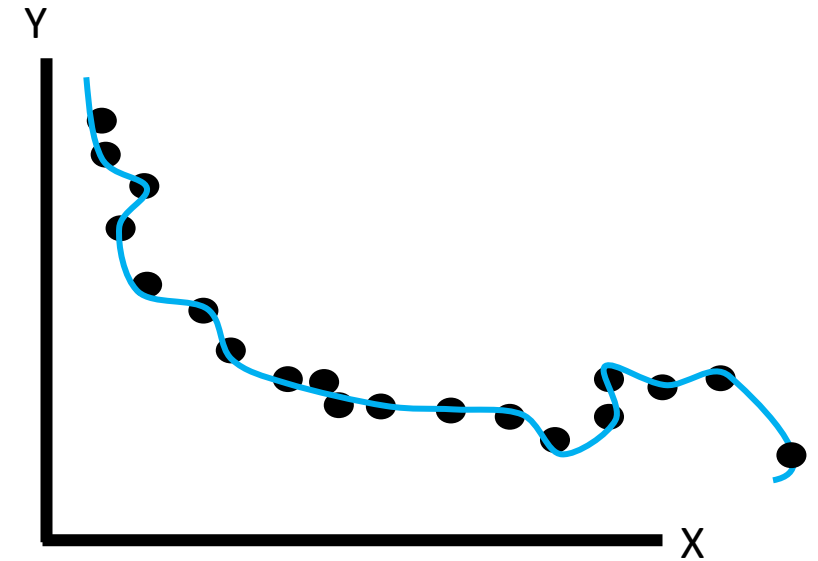
Under-fitting



Good fit



Over-fitting



Aiming for Generalization

Over-fitting

Training dataset is a **SAMPLE**

Why?

~100 items

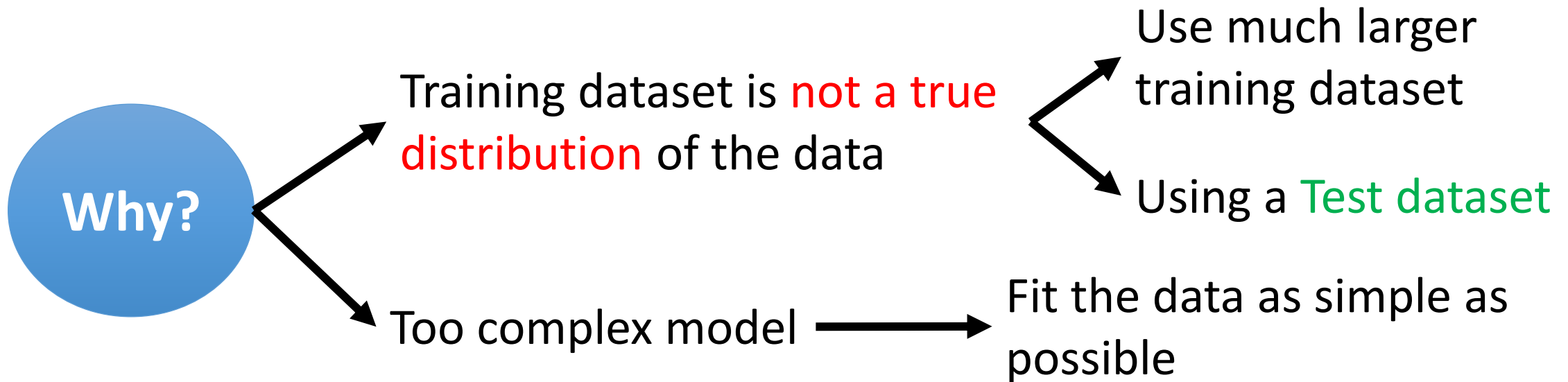


~1000/10K items



Aiming for Generalization

Over-fitting



Aiming for Generalization

Summary

1 Training a model is not an easy task...

2 Making a well-generalized model

3 Under-fitting and Over-fitting

Objective: making a good prediction on **NEW** data

